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A New Investigation of Wellbore Parameters Influence on Shut-In Pressure Transient Signatures from Water Hammer Fracture Diagnostics

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Abstract

Water hammer fracture diagnostics has been widely used in recent years, which enables field engineers to understand the stimulation effect from hydraulic fracturing treatment and optimize the completion strategy in time. Based on the high frequency acquisition equipment installed at the wellhead, the sampled water hammer signal is processed to obtain the reflection time of the fracture and the water hammer celerity in the wellbore, which is utilized to locate the downhole stimulation location. However, only the reflection time identified from the water hammer signal is used in current application, which means the waste of the useful information contained in the sampled signal.

In addition, the uncertainty of the water hammer celerity may lead to misjudgment of the downhole fracture, which is still challenging in nowadays diagnosis. Thus, profound investigation of the water hammer reflection characteristics and its utilization to accurately identify the information of downhole events is the key. Water hammer travels through the wellbore, carrying information about wellbore parameters, which are easy to obtain in the fracturing design or during the treatment. Studying the effect of wellbore parameters helps better understand the details of the water hammer, which helps accurately identify the fracture information. In this paper, the influence of the wellbore parameters on the water hammer reflection details is studied for the first time based on the numerical study, including the wellbore cementing and diameter change. Besides, water hammer signals from field cases are also studied, and the waveform characteristics related with wellbore parameters influence was analyzed, providing some useful guidelines for field application. The results of our study can improve the identification accuracy and provide useful insight for water hammer diagnostics.

Introduction

Multistage fracturing in horizontal well is an important development method for unconventional resources such as tight oil and shale gas (Yong et al. 2021; Jamaloei et al. 2022). However, due to formation heterogeneity and the failure of the bridge plug or ball during hydraulic fracturing treatment, not all perforating clusters contribute to production, which lead to the waste of construction resources (Haustveit

et al. 2017; Parkhonyuk et al., 2018a; Weddle et al., 2018; Ma et al. 2020). Fracturing diagnosis techniques help provide fracturing stimulation evaluation, guide the on-site completion decisions, and ensure the effect of fracturing treatment, which has been widely developed and applied in recent years (Bhatnagar et al., 2016; Watson et al., 2022; Leggett et al., 2023; Olson et al., 2023; Shokanov et al., 2023).

Water hammer diagnostics, as a real-time, simple operation, low-cost and non-intrusive fracturing diagnosis technique, has been applied since 2016 to evaluate downhole events such as downhole fluid acceptance location, temporary plug effect, bridge mechanical isolation effect and case leaking in unconventional oil and gas hydraulic fracturing (Bogdan et al., 2016). The schematic diagram of the water hammer diagnostics in hydraulic fracturing treatment is shown in Figure 1. The key of locating downhole events is the determination of the reflection time between wellhead and downhole fracture and the propagation speed of water hammer in the wellbore (Parkhonyuk et al., 2018b; Korkin et al., 2020). The pressure transient observed at the end of injection, known as water hammer, is generated at the wellhead and propagates through the wellbore. While encountering the downhole events, the wave reflects back to the surface and is collected by the equipment installed at the wellhead. Therefore, the reflection time obtained from the water hammer contains information about the characteristic depth of the wellbore and the fracture. Through the signal processing model based on cepstrum analysis, the response time of water hammer reflection between wellhead and downhole fracture can be obtained.

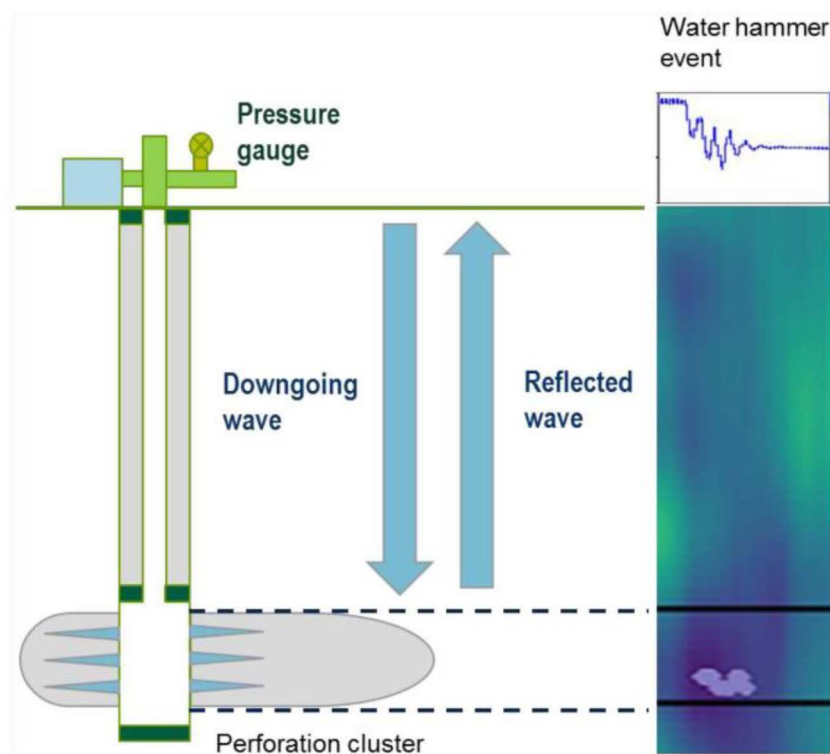


Figure 1—the schematic diagram of the water hammer diagnostics (Panjaitan et al., 2018).

The water hammer celerity depends on several parameters, including the pressure, temperature, borehole geometry and fluid properties (Rutzing et al., 2017). The typical celerity value varies from 4,000 and 5,500 ft/s (about 1219 to 1676 m/s). It is challenging to accurately calculate the speed of the water hammer travel through the wellbore and decrease the water hammer celerity uncertainty. Large variation range will lead to huge error in water hammer diagnosis results. At present, the determination of water hammer celerity in water hammer diagnostics is based on Bayesian algorithm (Kabannik et al., 2018; Parkhonyuk et al., 2018b;

Korkin et al., 2020). Finally, combined with the response time of fracture reflection and the celerity of water hammer pressure wave, the location of downhole fluid entrance point is identified.

However, only the reflection time identified from the water hammer signal is used in current application, which means the waste of downhole information carried in the water hammer exists and the limitation of the evaluation effect in water hammer diagnostics (Iriarte et al. 2017). In addition, due to the uncertainty of the water hammer celerity, the inaccurate understanding of the water hammer wave propagation speed in the wellbore may lead to the misjudgment of the downhole fracture. Therefore, it is very important to deeply study the reflection characteristics in the water hammer waveform and utilize the wave reflection characteristics and the known information to enhance and accurately understand the information of downhole events. The wellbore parameters are easy to obtain in the fracturing design or during the treatment. Water hammer travels through the wellbore, carrying information about wellbore parameters. Thus, studying the effect of wellbore parameters helps better understand the details of the water hammer, which helps accurately identify the fracture information

In this paper, a water hammer simulator based on fluid transient model is developed, and the influence of the wellbore parameters on the water hammer reflection details is studied for the first time, including the wellbore cementing and diameter change. Besides, water hammer signals from field cases are also studied, and the waveform characteristics related with wellbore parameters influence was analyzed, providing some useful guidelines for the water hammer diagnostics.

Water hammer simulation model

Transient flow model. After pumps shut-in, the water hammer is generated at the wellhead. At the end of the fracturing, with the rapid reduction of the flow rate at the surface, the water hammer is generated at the wellhead and propagates through the wellbore. When the water hammer encounters the downhole boundary (the fracture or other condition), it reflects and propagates back to the wellhead, collected at the wellhead by pressure transducer (Liang et al., 2017). This pressure transient in the wellbore can be modeled through the equations of motion and continuity (Carey et al., 2015; Ma et al., 2019),

$$\frac{\partial H}{\partial t} + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0 \quad (1)$$

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{fQ|Q|}{2DA} = 0 \quad (2)$$

where a is the water hammer celerity in the wellbore, A is the wellbore cross-sectional area, H is the pressure head, Q is the flow rate in the pipeline, D is the wellbore diameter, and f is the wellbore friction coefficient, t is the time, x is the measured depth along the well.

The equation (1) and (2) can be converted from two partial differential equations to four ordinary differential equations by the method of characteristic. The arranged characteristic line equations are as follows:

$$\begin{cases} \frac{dQ}{dt} \pm \frac{gA}{a} \frac{dH}{dt} + \frac{f}{2DA} Q|Q| = 0 \\ \frac{dx}{dt} = \pm a \end{cases} \quad (3)$$

Using these equations and the known value at the time t . The pressure head H and flow rate Q at the time $t+\Delta t$ at the intersection of characteristic lines can be obtained by the integral of differential equations, which can be solved in the form of finite difference, and the second order approximation is used to sort out the formula and integrate it,

$$C^+ : H_P = H_A - B(Q_P - Q_A) - RQ_P|Q_A| \quad (4)$$

$$C^- : H_P = H_B + B(Q_P - Q_B) + RQ_P|Q_B| \quad (5)$$

where B and R represent the pipe characteristic impedance and the resistance coefficient of the pipe respectively,

$$B = \frac{a}{gA} \quad (6)$$

$$R = \frac{f\Delta x}{2gDA^2} \quad (7)$$

Under initial conditions, H and Q are known for each position i at the wellbore. The H and Q after time Δt can be calculated from the known values at position $i-1$ and $i+1$ at time t . Therefore, the finite difference form can be written as:

$$C^+ : H_i^{t+\Delta t} = C_P - B_P Q_i^{t+\Delta t} \quad (8)$$

$$C^- : H_i^{t+\Delta t} = C_M + B_M Q_i^{t+\Delta t} \quad (9)$$

The H and Q of each position in each time step can be solved by equation (8) and (9),

$$H_i^{t+\Delta t} = \frac{C_P B_M + C_M B_P}{B_M + B_P} \quad (10)$$

$$Q_i^{t+\Delta t} = \frac{C_P - C_M}{B_M + B_P} \quad (11)$$

where C_P , B_P , C_M and B_M are

$$C_P = H_{i-1}^t + B Q_{i-1}^t \quad (12)$$

$$B_P = B + R |Q_{i-1}^t| \quad (13)$$

$$C_M = H_{i+1}^t - B Q_{i+1}^t, \quad (14)$$

$$B_M = B + R |Q_{i+1}^t| \quad (14)$$

Boundary conditions. To solve the propagation of water hammer wave in wellbore, the boundary conditions specification is needed. In this model, the fluid injected at the wellhead is assumed at the rate of Q_0 , which is linearly decreased to 0 in a very short time. The outlet boundary of the model is set as the constant pressure boundary. The existence of fractures communicated the formation pressure system and the well bottom (Carey et al. 2015; Dung et al., 2021). However, this paper mainly studies the propagation behavior in the wellbore and the influence of wellbore parameters, which simplifies the setting of bottom hole condition.

Wellbore parameters configuration

Wellbore cementing. During the completion process, cement outside the casing is not returned to the surface. The water hammer celerity in the wellbore can be calculated by (KLEM et al., 2017),

$$a = \sqrt{1 / \left(\frac{\omega}{g} \left(\frac{1}{K} + \frac{dc}{Ee} \right) \right)} \quad (16)$$

$$c = \begin{cases} 5/4 - \mu & \text{Casing cemented at upper end only} \\ 1 - \mu^2 & \text{Casing cemented at upper and bottom} \\ Ee / (Gd + Ee) & \text{Casing cemented uniformly} \end{cases} \quad (17)$$

where a is the water hammer celerity traveling in the wellbore, ω is the specific gravity of the fluid, g is the acceleration of gravity, K is the volume modulus of fluid, d is the pipe diameter, c is cementing coefficient, E is the Young's modulus of the pipeline, e is the pipe thickness, μ is the Poisson ratio of pipe, G is the shear modulus of rock.

The water hammer celerity in the wellbore with or without cementing can be described by the cementing coefficient calculated by [equation \(17\)](#), where the celerity will change significantly. Thus, the influence of wellbore cementing on the wellhead water hammer can be simulated by adjusting the celerity in different wellbore segments in the model.

Wellbore diameter variation. When it comes to the wellbore diameter variation, the wellbore can be assumed as two pipe section with different diameter, as pipe j and pipe k shown in [Figure 2](#).

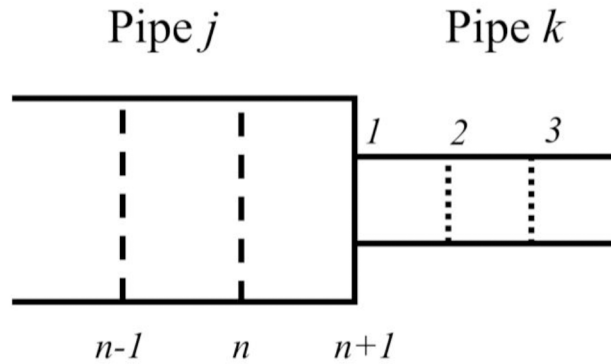


Figure 2—the grid connection schematic of the wellbore diameter variation section.

The additional configuration is introduced to solve the H and Q at the end of pipe j. The boundary condition will be solved by the continuity [equation \(18\)](#) and [\(19\)](#), where it is assumed that the head loss at the pipe connection is ignored. By substituting [equation \(18\)](#) and [\(19\)](#) into [equation \(8\)](#) and [\(9\)](#), the solution of wellbore section j and k can obtain H and Q of the diameter variation position.

$$H_{j,n+1}^t = H_{k,1}^t \quad (18)$$

$$Q_{j,n+1}^t = Q_{k,1}^t \quad (19)$$

Model validation. First, we used the experimental data of [Hu et al. \(2022\)](#). A 156-m-long steel pipe with diameter of 0.025m for the experiment was installed ([Hu et al., 2022](#)). The inlet of the pipeline was connected with a continuously variable speed pump to provide continuous flow in the long pipeline. The sensor was located 40 m away from the upstream end. The pipe outlet provided the pressure boundary through the connecting of the pressure vessel. The flow rate before the pump shutdown was 0.033m³/min, and the shutdown time of the pump was 6 second. The measured pressure response at a distance of 40 m upstream was compared with the numerical results simulated by our model in this paper, as shown in [Figure 3](#). The numerical results are in good agreement with the experimental data, and the wave reflection characteristic has been accurately captured.

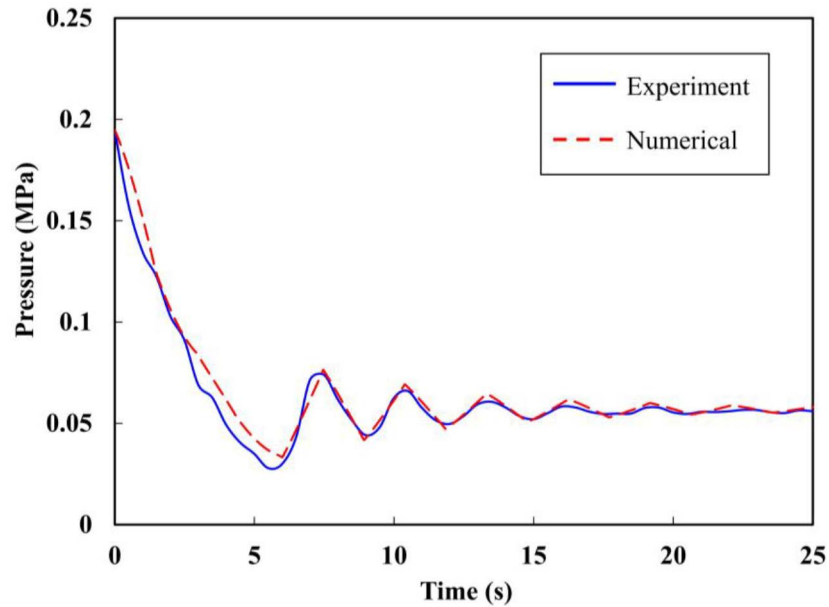


Figure 3—Comparison of the numerical results and experimental sampled data in a uniformly long pipeline.

Second, experimental data of Gong et al.(2014) were used to verify the water hammer transient in the pipe with locally enlarged diameter (Zhang et al., 2021). The length of the pipe was 37.46m with the inner diameter is 22.14 mm. The total length of the pipeline with the enlarged diameter 22.96 mm was 1.649 mm, which was located at 17.805 m upstream of the valve. The valve quickly closed. The comparison between numerical and experimental results was shown in Figure 4, which was observed in good consistent.

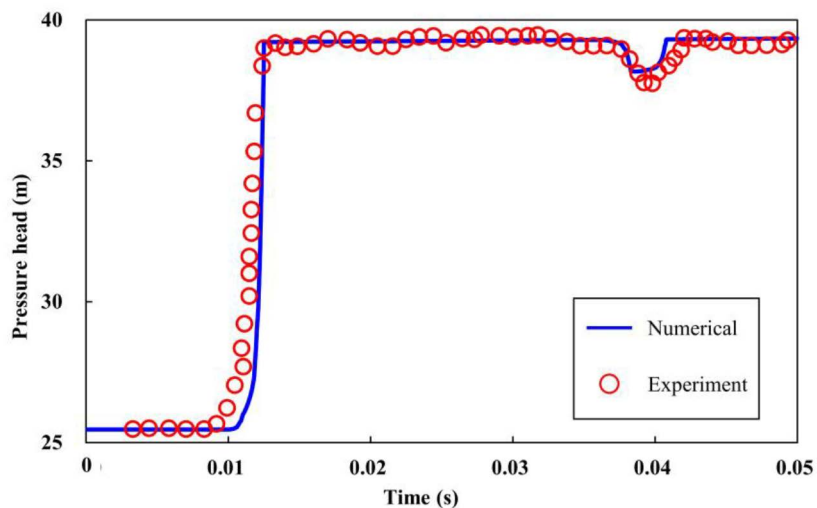


Figure 4—Comparison of the numerical and experimental results in the pipe with locally enlarged diameter.

Simulation results and discussion

Water hammer reflection characteristic. Figure 5(a) displays a result of the wellhead water hammer simulated by our water hammer model in a uniformly wellbore, indicating that water hammer signatures can reproduce the main features observed in the field data. The parameter values are $\rho=1000 \text{ kg/m}^3$, water hammer celerity in the wellbore $a=1600 \text{ m/s}$, and wellbore length $L=3000 \text{ m}$. The initial injection rate $Q_0=3 \text{ m}^3/\text{min}$, and the time of the inlet flow rate decreasing to zero in 0.8 s. Figure 5(b) shows the 3 reflection features of reflection characteristics in the wellhead water hammer propagating through the wellbore: (i)

as the injection rate decreases, the pressure drop at the wellhead occurs immediately, (ii) due to the line pack effect, the water hammer waveform shows a slow trend of pressure decreasing, rather than a horizontal platform (no pressure change), and (iii) the first fracture reflection at the bottom traveling back to the surface and the rapid pressure increase occurs.

The pressure drop in the reflection feature (i) of the wellhead water hammer is caused by the decrease of the injection rate. meanwhile, effective information about the fracture only appears in the wellhead water hammer signal after the first fracture reflection (in the reflection feature (iii)). Thus, the pure influence of wellbore parameters on wellhead water hammer is concentrated in the reflection feature (ii) of the water hammer reflection characteristics. The detailed investigation in the reflection feature (ii) is the key to study the influence of wellbore cementing and its diameter variation on wellhead water hammer signature details.

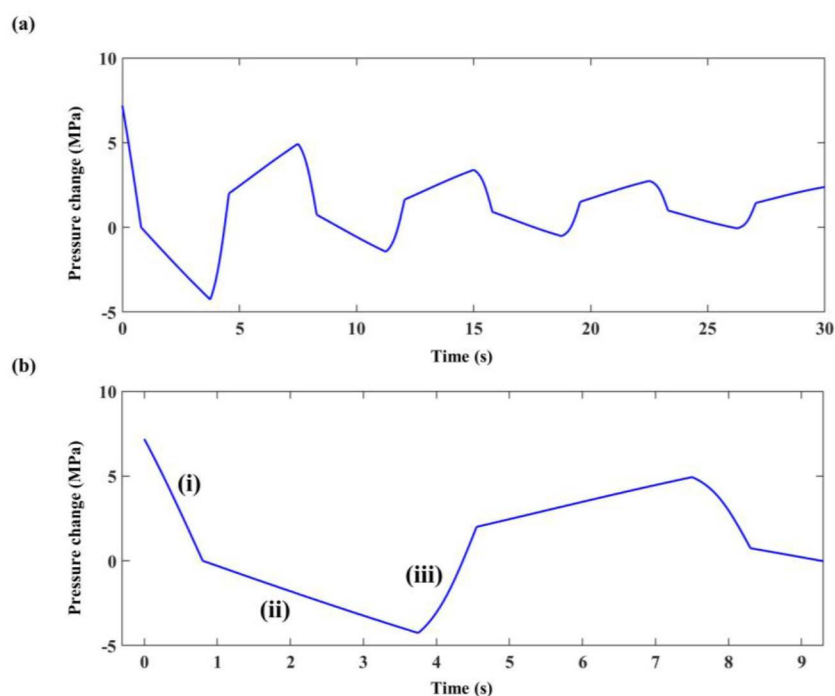


Figure 5—(a) the overall wellhead water hammer simulation result in a uniformly wellbore. (b) the zoomed in water hammer result to illustrate the reflection characteristics.

The effect of the wellbore cementing. Water hammer wave has different reflection characteristics under different properties of pipes. In this section, 4 cases are designed to analyze the wellhead water hammer response characteristics. The wellbore schematic of uncemented and cemented section are shown in Figure 6.

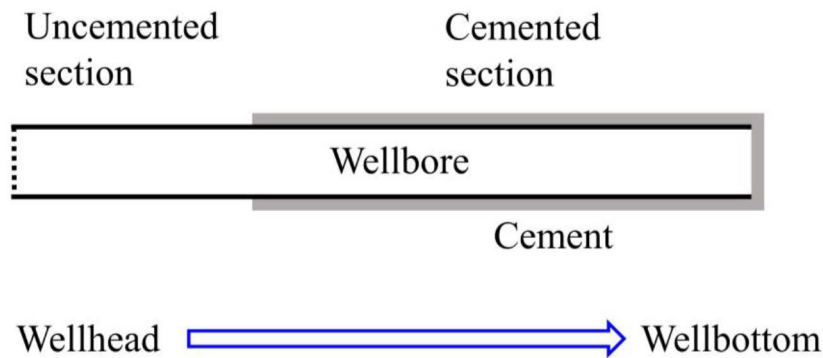


Figure 6—the wellbore schematic of uncemented and cemented section.

The simulation results are shown in Figure 7(a). In the reflection feature (ii) of the water hammer, the pressure trend is the same in the uncemented and fully cemented wellbore. This is because the decline of this section is affected by the wellbore friction, which is the embodiment of the line pack effect (Walters et al. 2019). For the partially cemented wellbore, the cementing section accelerates the original pressure drop trend, which leads to a pressure trend change in the reflection feature (ii) of the water hammer. The pressure trend change occurs at the time 2.857 s and 1.428 s, respectively. Combining the water hammer celerity, the location calculated by these response times is about 2000 and 1000 m, indicating the length of the uncemented wellbore. The duration time of the pressure trend change is 0.8 s, which is consistent with the pump shut-in time. As the length of the cementing section becomes longer, the reflection of the fracture appears more quickly, which indicating the average wave velocity inside the wellbore increases. In our cases, the fracture reflection time of the fully cemented wellbore is 0.535 s earlier than that of the uncemented wellbore, and the water hammer celerity of the cementing section is about 14% higher than that of the uncemented section. Wellbore cementing can significantly increase the speed of water hammer wave propagation in the wellbore, which is crucial for accurate diagnosis of downhole fracture location. Meanwhile, the peak and valley value of wellhead water hammer wave increases gradually with the increase of cementing section length. The energy dissipation of the water hammer in the cementing wellbore is smaller, and the amplitude of the water hammer is larger. With the reflection of the water hammer in the wellbore (after the reflection feature (iii)), the wellhead water hammer becomes more complex in partial cemented wells.

The effect of the wellbore diameter variation size. In this section, the effect of the wellbore diameter variation size on the response characteristics of wellhead water hammer wave is discussed with 5 simulation cases. The diameter variation position is located at the 2500 m in the fully cementing wellbore with the 50-m-long diameter variation section. The variation diameters are 0.08, 0.1, 0.125, 0.15 and 0.2 m, respectively. The results are shown in the Figure 7(b). The wellbore with smaller diameter will accelerate the decline trend of the pressure in the wellhead water hammer, while the wellbore with the enlarged diameter slows down the pressure drop tendency. The reason of this phenomenon is the change of the wellbore diameter alters the impedance at the corresponding position, where the water hammer wave reflects. The positive or negative reflection coefficient determines the pressure change in the reflection feature (ii) of the water hammer. The reflection coefficient is calculated as follows (Dunham et al. 2023):

$$R = \frac{Z - Z_w}{Z + Z_w} \quad (20)$$

where Z is the impedance of the wellbore diameter variation section, Z_w is the characteristic impedance of the wellbore.

When the wellbore diameter decreases, the impedance of the corresponding position Z becomes larger, which is greater than the characteristic impedance of the wellbore Z_w . The reflection coefficient is positive, and positive reflection occurs, which accelerates the pressure decline trend in the reflection feature (ii) of the water hammer. As the wellbore diameter increases, the impedance of the corresponding position Z becomes smaller, which is less than the characteristic impedance of the wellbore Z_w . The reflection coefficient is negative, and negative reflection occurs, which deaccelerate the pressure decline trend in the reflection feature (ii) of the water hammer. The magnitude of the pressure change is also related to the size of the wellbore diameter variation. The greater the diameter change, the larger the water hammer pressure change.

The effect of the wellbore diameter variation position. The effect of the wellbore diameter variation position on the wellhead water hammer wave is discussed with 4 cases in this section. The variation diameter is 0.15 m (enlarged diameter) with the length of 50 m. They are located at 1000m, 1500m, 2000m and 2500m positions of the fully cementing wellbore, respectively. The results are shown in Figure 7(c). In the reflection feature (ii) of the water hammer, negative pressure reflection appears successively. The time of occurrence is 1.25, 1.875, 2.5 and 3.125 s respectively. According to the water hammer wave celerity set in our model, the corresponding enlarged diameter positions are indicated, respectively. The negative pressure reflection duration was the same in each case (0.063 s in this paper), which was related to the length of the enlarged diameter section. It is worth noting that during a certain time after the first reflection representing the wellbore diameter variation, the pressure converted into the initial trend (0.8 s in this paper), which is consistent with the time that injection rate becomes to zero at the wellhead.

The effect of the wellbore diameter variation length. This section discusses the effect of the wellbore diameter variation length on the wellhead water hammer wave in 5 cases. The diameter variation position is located at the 2500 m in the fully cementing wellbore with the 0.1 m diameter (reduced diameter). The wellbore diameter variation length are 20, 40, 60, 80, and 100 m, respectively. The results are shown in Figure 7(d). As the length increases, the positive reflection duration increases in the reflection feature (ii) of the wellhead water hammer. At the same time, with the increase of reduced diameter section, the average celerity in the wellbore increases, and the reflection of the downhole fracture becomes earlier gradually.

The numerical simulation results show that the wellbore cementing accelerates the change of pressure trend and changes the average water hammer celerity in the wellbore. The moment pressure trend alters depends on the length of the wellbore cementing. The pressure amplitude of pipe diameter change is affected by the degree of pipe diameter change and the length of pipe diameter change. Wellbore diameter reduction will show negative reflected pressure increase in the details of the water hammer waveform, while diameter expansion will show positive reflected pressure subsidence. The position of reflection is related to the position of reducing diameter, and the duration of reflection is related to the length of reducing section. The reflection characteristics of wellhead water hammer wave will become more complex whether it is cementing or partial diameter change in the wellbore. Meanwhile, since these effects are caused by the change of wellbore parameters, these characteristics will first be reflected in the reflection feature (ii) of the wellhead water hammer, which takes priority before the reflection feature (iii) of the downhole fracture, guiding us to process and analyze the complex water hammer signals collected on site.

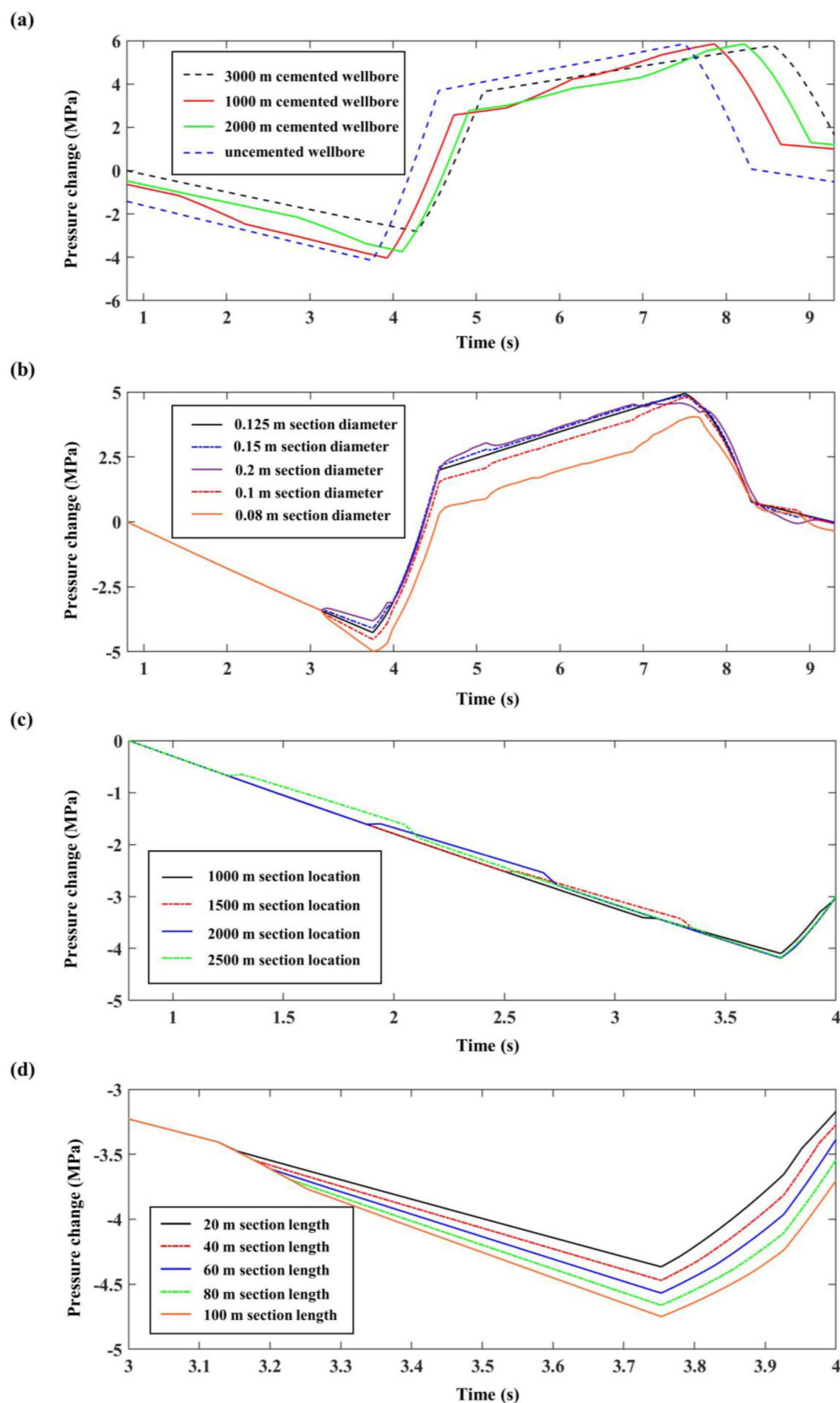


Figure 7—(a) the simulation wellhead water hammer results under different wellbore cementing condition. (b) the simulation wellhead water hammer results with different wellbore diameter variation size. (c) the simulation wellhead water hammer results with different wellbore diameter variation position. (d) the simulation wellhead water hammer results with different wellbore diameter variation length.

Field data investigation and analysis

The effect of the wellbore cementing. The first case displayed the effect of the wellbore cementing on the pump shut-in water hammer signal sampled in well A, which was a typical partially cemented wellbore fracturing well. The wellbore structure of well A is shown in Figure 8(a). The fracturing casing length was 5028 m and the cement return height is 2950 m. The fracturing interval of this stage was 4300 to 4375 m. The length of the cemented section was 2193 m and the uncemented section is 2182 m. According to the DAS interpretation of this well, the fluid acceptance location was 4345 m. According to the cepstrum analysis, the fracture response time of this section was 6.021 s, and the average water hammer celerity in the wellbore was about 1455 m/s. In the reflection feature (ii) of the wellhead water hammer, there was a clear trend of increasing pressure drop shown in red and green arrow in Figure 8(b). The time when the pressure drop trend increased was about 3.2 s. According to the information of the wellbore structure, the wave velocity in the cementing section increased by about 14.45 % compared with that in the uncemented section. It had a good agreement to the results obtained in the previous study of our numerical model.

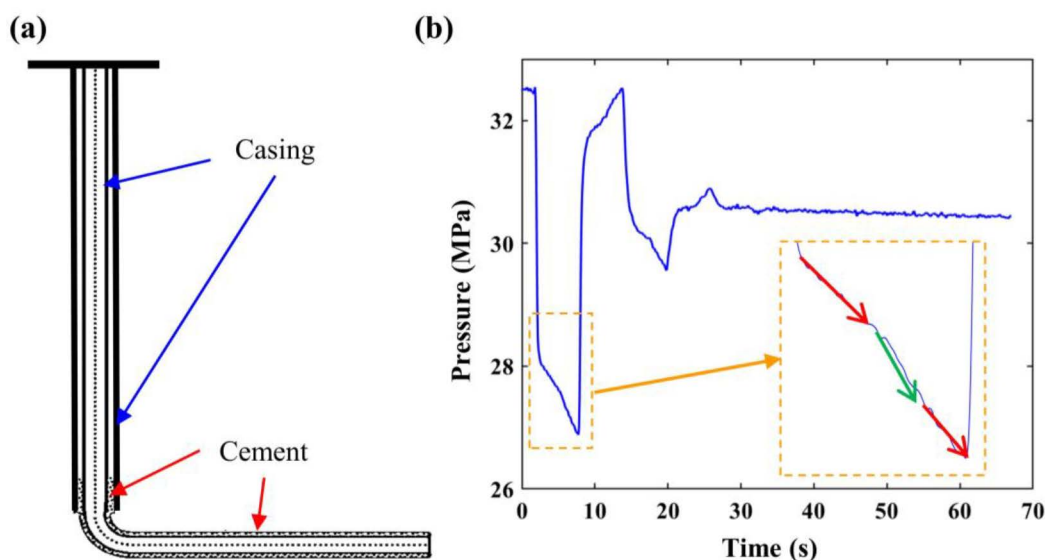


Figure 8—(a) the wellbore structure of the well A. (b) the pump shut-in wellhead water hammer signal sampled from well A during hydraulic fracturing treatment.

Reflection time variation during hydraulic fracturing (wellbore cementing quality impact). The second example showed the effect of cementing quality on fracture reflection time obtained from the water hammer diagnostics. Wells A and B were multistage fracturing wells designed in perf-and-plug completion method in tight reservoirs. the statistical results illustrated that the overall tendency of the fracture reflection time gradually decreased with the progress of fracturing, which was consistent with the sequence of fracturing from toe to heel of the wellbore. According to the results of DAS monitoring or other methods, there was no indication of plug isolation failure, and the stimulation was successfully in the target stage. Under the circumstance, the measured depth of the target zone decreased, and the reflection time appeared in the decline trend. However, some reflection times in both well A and B were not following that trend. For example, in the case of well A in Figure 9 (a), fracture reflection time increased in stage 2 to 6 and stage 11 to 13. According to the previous theoretical study, the presence of cementing section altered the propagation speed of water hammer in the wellbore. Thus, the cementing quality would also affect the velocity of water hammer wave in the wellbore to different degrees. In well A, wellbore logging results showed bad cementing quality from sections 2 to 6 and 11 to 13 at corresponding measured depth. Similar regularities were also found according to the statistical results of fracture reflection time of well B in Figure 9(b). The cementing quality would change the decline trend of fracture reflection time in the process of

fracturing. Whether to slow down or convert to an increasing trend was related to the specific cementing quality. How to fully consider the effect of the wellbore cementing quality in the water hammer celerity prediction model in water hammer diagnostics is one of the keys to improve the final diagnosis precision.

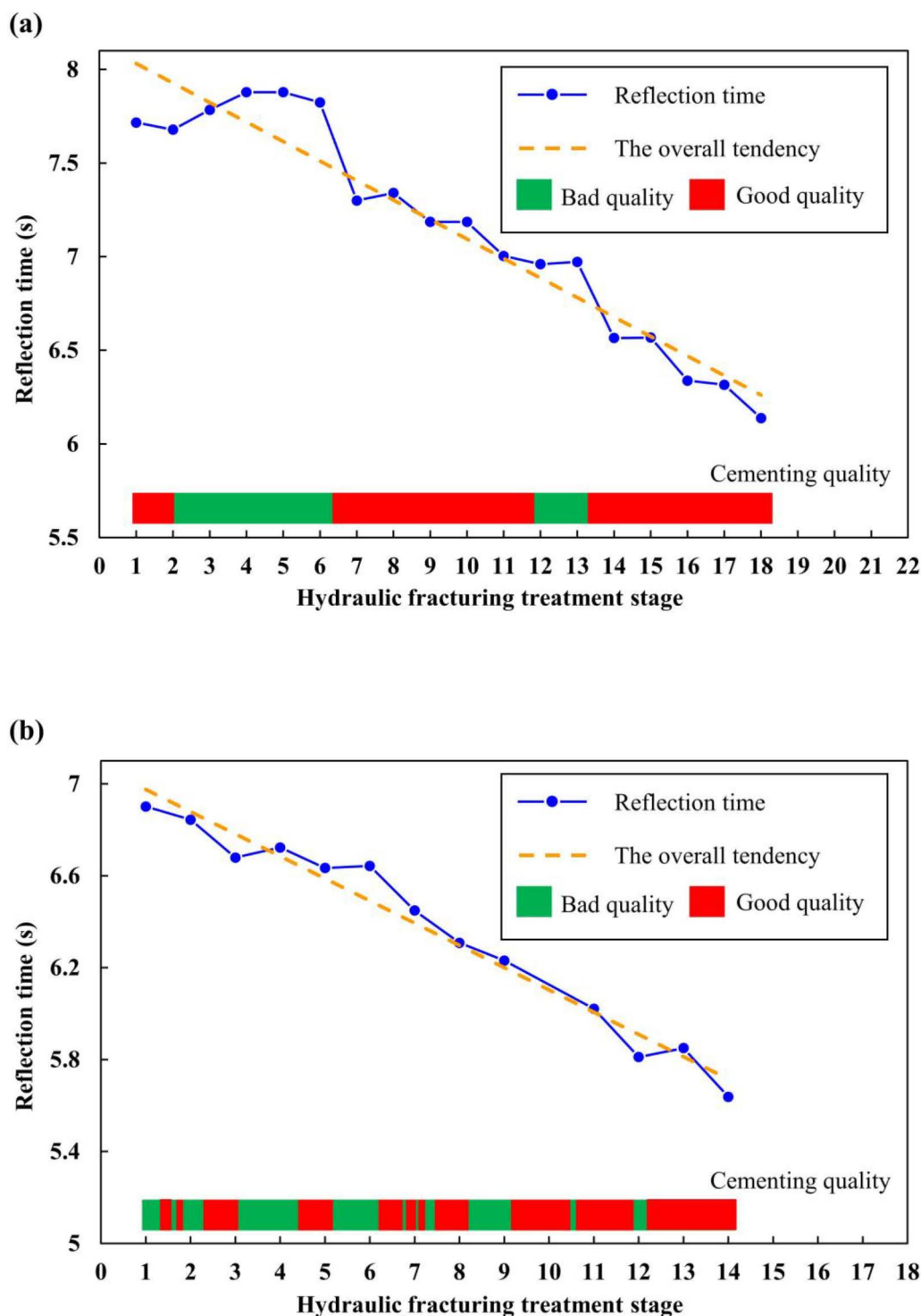


Figure 9—(a) the fracture reflection time statistical results and the cementing quality of the well A. (b) the fracture reflection time statistical results and the cementing quality of the well B.

The effect wellbore diameter variation. In the third example, we discussed the effect of wellbore diameter variation on the wellhead water hammer. Well structure diagram of well C was shown in Figure

10(a). The first section was 3-1/2" tubing with the length of 6030 m. The second section was 7-5/8" cementing casing with the length of 1310 m, and the third section was 5" uncemented casing hanging from the bottom of the second section with the length of 2056 m. The blue part represented the downhole packers. The wellhead water hammer of well C at the end of the fracturing was shown in Figure 10(b).

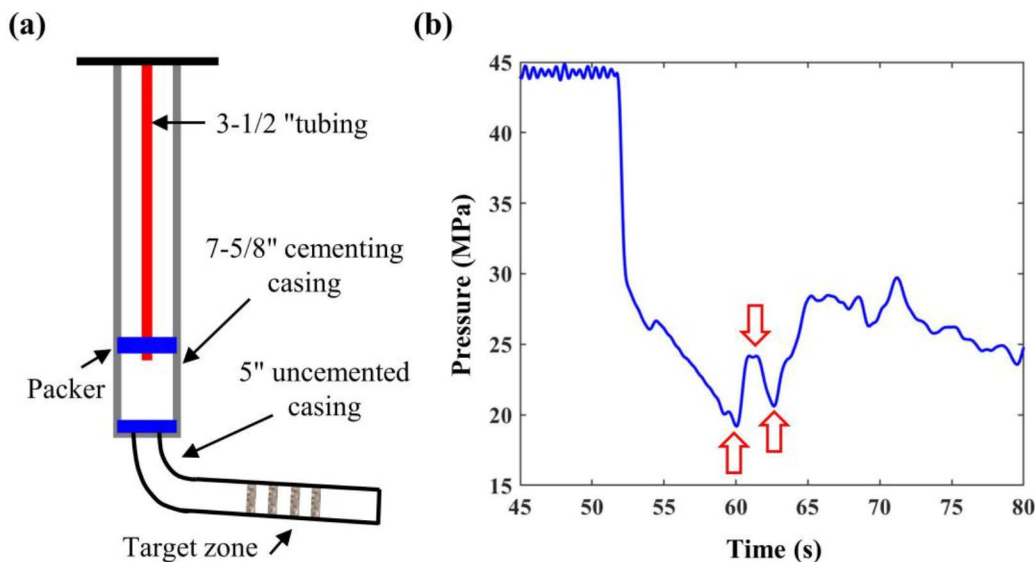


Figure 10—(a) the wellbore structure of the well C. (b) the pump shut-in wellhead water hammer signal sampled from well B during hydraulic fracturing treatment with labeled arrow.

The red arrows were marked as the points where significant pressure reflection occurred, corresponding to the wellbore diameter variation in the well structure. The first reflection point was a negative reflection, corresponding to the enlarged wellbore diameter. The second reflection was positive and corresponded to the reduction of the wellbore diameter. The time of pressure reflection was 8.28 s and 9.705 s after pump shutdown, respectively. According to the wellbore diameter variation location and pressure reflection time, the average celerity corresponding to different wellbore section could be obtained, which was 1462 m/s and 1512 m/s. Although the diameter of the tubing was small, it was suspended through the wellhead and the downhole packer and not cemented. the water hammer wave propagated in the tubing was less than the velocity in the large diameter cementing casing. The cepstrum of the water hammer signal of well C is shown in the Figure 11. Multiple responses in the cepstrum existed, representing not only the wellbore diameter variation. Misjudging the impact of the well structure in the wellhead water hammer would lead to misdiagnose of the downhole fluid acceptance location.

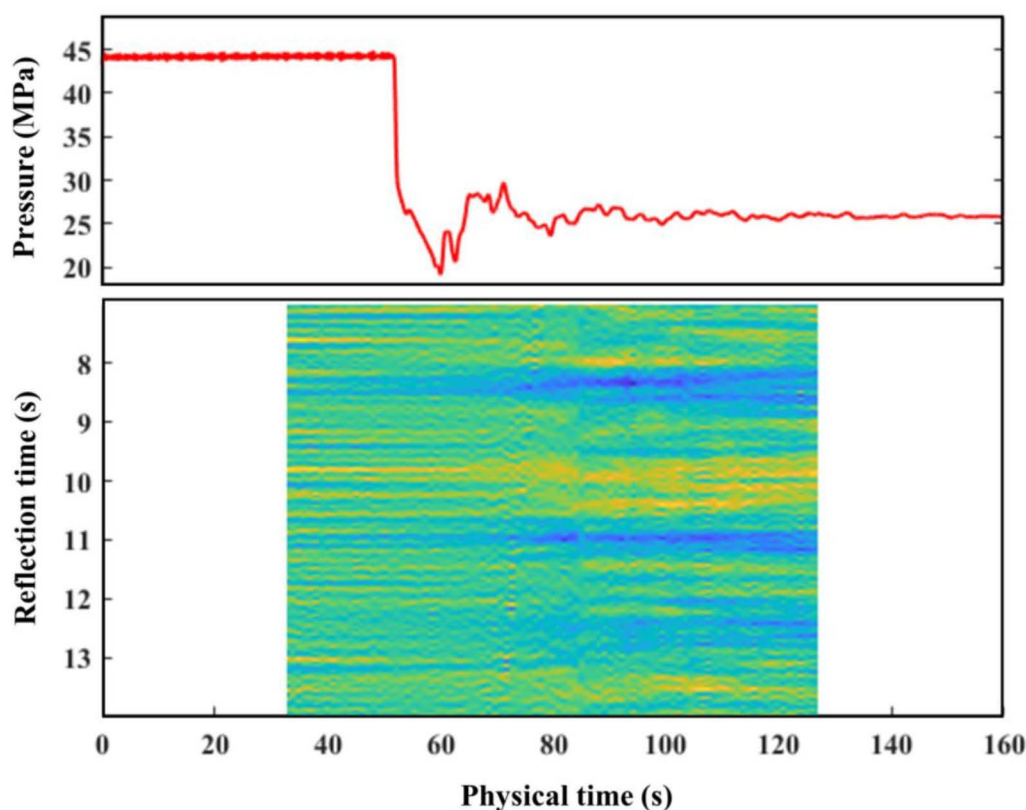


Figure 11—The cepstrum analysis of the water hammer signal sampled from well C (the response in blue is the negative reflection indicator).

Conclusion

In this paper, the effect of the wellbore parameters on the wellhead water hammer were studied for the first time based on the wellbore transient flow model establishment. In addition, water hammer signals from field cases are also studied, and the waveform characteristics related with wellbore parameters influence was analyzed, providing some useful guidelines for the water hammer diagnostics. First, due to the line pack effect of the wellbore, a slow change trend of the water hammer waveform is showed rather than a horizontal one between the time of pump shut-in and the downhole boundary reflection. The wellbore cementing accelerates the change of the pressure trend, and the moment it occurs in the waveform depends on the length of the wellbore cementing. The wellbore diameter reduction will appear as a pressure spike on the water hammer waveform detail, while the diameter expansion will appear as a pressure sinking. The pressure amplitude of the diameter change is influenced by the diameter variation degree and the length of the wellbore diameter change. Second, the water hammer waveform shows that the reflection of the fracture occurs earlier due to the existence of the cemented wellbore section in numerical study. The water hammer celerity in cemented section is about 14 % larger in uncemented section in field case investigation. Third, the water hammer reflection time is negatively correlated with the fracturing treatment stages. However, bad quality of the partial wellbore cementing will alter the water hammer reflection time tendency, which may lead to misidentification of the downhole fluid acceptance location. The information about the length of the wellbore cementing and the diameter variation can be the indicators that help calibrate the water hammer celerity, which is variable during the fracturing stages and the key part in the water hammer diagnostics.

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